

DSD Final Project Scores

1. BrPred

(1) Total execution cycles of given I_mem_BrPred: 566

截圖：

```
[INFO]: Testing with pattern <BrPred>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====
Reset Done ...
=====

      .##.
     .##'##'
    .##L##
   .##P##
  '##L##'
   '##'
   .##.
  .##' ##.
 .##'  ##.
.##'   ##.
:
:
:
##      ##

=====
\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!
=====

[Performance Statistics]
- Total Cycle Count   : 566
- D-Cache Stall Cycles: 101
- I-Cache Stall Cycles: 147
- Data Memory Reads   : 1
- Data Memory Writes  : 1
- Inst Memory Reads   : 8
- Branch Instructions  : 79
- Branch Prediction Successes : 58
- Branch Prediction Failures : 21
- Branch Prediction Rate : 73%
=====
```

(2) Total execution cycles of given I_mem_hasHazard: 2698

截圖：

```
[INFO]: Testing with pattern <hasHazard>
```

```
START!!! Simulation Start .....
```

```
=====
```

```
Reset ...
```

```
=====
```

```
=====
```

```
Reset Done ...
```

```
=====
```

```
Current cycle = 1000
```

```
Current cycle = 2000
```

```
      .##.
```

```
    .##'##'
```

```
  .###L###
```

```
.#####
```

```
'#####'
```

```
  #####
```

```
    .###.
```

```
  .###.
```

```
..#'.##.
```

```
..
```

```
.
```

```
:
```

```
..          ..
```

```
#l         #
```

```
=====
```

```
\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!
```

```
=====
```

```
[Performance Statistics]
```

```
- Total Cycle Count   : 2698
```

```
- D-Cache Stall Cycles: 242
```

```
- I-Cache Stall Cycles: 293
```

```
- Data Memory Reads   : 5
```

```
- Data Memory Writes  : 5
```

```
- Inst Memory Reads   : 16
```

```
- Branch Instructions  : 285
```

```
- Branch Prediction Successes : 137
```

```
- Branch Prediction Failures  : 148
```

```
- Branch Prediction Rate : 48%
```

```
=====
```

(3) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): (ns)

2.7

2. Compressed instructions

(1) Total Simulation Time of given I mem compression: (ns) 2112.405

截圖：

截圖：

(4) Total Simulation Time of LFSR_HIST: (ns) 710929.905

截圖：

$$(5) \ A \times T_{Q_sort} \times T_{Conv} \times T_{LFSR_HIST} : (um^2 * ns^3)$$

2.84062*10²¹

(6) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): (ns) 2.7